

Society and Technology

BCA — Semester 1 — 2018 — Answer Sheet

Subject Code: CACS102

Group B

Attempt any SIX questions. [6×5=30]

11. How sociology is a general science? Discuss with examples.

[5]

Answer: Sociology as a General Science:

Sociology is considered a general science because it studies society and social behavior using systematic, objective, and scientific methods rather than relying on speculation or tradition.

Reasons why sociology is a general science:

1. **Scientific Method:** Sociology employs observation, hypothesis formulation, data collection, and analysis — the same scientific methodology used in natural sciences. For example, Emile Durkheim's study of suicide used statistical analysis to establish correlations between suicide rates and social integration.
2. **Objectivity:** Sociologists strive to study social phenomena without personal bias. Max Weber advocated "value-free" sociology, where researchers separate personal values from scientific investigation.
3. **Universal Laws:** Sociology seeks to discover general principles about human social behavior that apply across different societies. For instance, the principle that industrialization leads to urbanization has been observed globally.
4. **Empirical Verification:** Sociological theories can be tested against observable reality. The theory of social stratification can be verified by examining income distribution data in various countries.
5. **Systematic Study:** Sociology does not study society randomly; it has systematic theories (functionalism, conflict theory, symbolic interactionism) that organize knowledge about social life.

12. "Family is a primary social institution". Discuss with suitable examples.

[5]

Answer: Family as a Primary Social Institution:

A **primary social institution** is one that is fundamental to society, universal across cultures, and essential for social stability and continuity. Family qualifies as the most primary of all institutions.

Why family is a primary institution:

1. **Universal Presence:** Every known society has some form of family structure. Whether nuclear, extended, or joint, the family exists everywhere as the basic unit of social organization.
2. **Biological Reproduction:** Family is the institution through which society reproduces itself biologically. It regulates sexual behavior and ensures the continuation of the human species through legitimate childbearing.
3. **Socialization of Children:** Family is the first and most important agent of socialization. Children learn language, norms, values, and culture within the family. For example, a child learns to speak Nepali, respect elders, and celebrate Dashain primarily through family influence.
4. **Economic Cooperation:** Historically, family served as the basic economic unit where members cooperated for survival. In agrarian societies, family farms were the primary production units. Even today, family businesses remain significant economic entities.
5. **Emotional Support:** Family provides emotional security, love, and care that no other institution can fully replace. This emotional foundation is essential for individual mental health and social stability.
6. **Regulation of Social Roles:** Family assigns roles (father, mother, child, sibling) that prepare individuals for broader social roles. A child who learns to cooperate with siblings is better prepared for workplace teamwork.

13. Define socialization and describe its stages and agents.

[5]

Answer: Socialization:

Socialization is the lifelong process through which individuals learn the culture, norms, values, and behaviors appropriate to their society. It transforms a biological organism into a social being capable of participating in social life.

Stages of Socialization:

1. **Primary Socialization (0-5 years):** Occurs within the family. Children learn basic language, motor skills, emotional control, and fundamental values. This is the most formative stage.
2. **Secondary Socialization (6- adolescence):** Occurs in schools, peer groups, and through media. Children learn specific skills, academic knowledge, and broader social norms beyond the family.
3. **Adult Socialization:** Occurs in workplaces, marriage, and parenthood. Adults learn occupational roles, marital expectations, and parenting skills. This stage involves resocialization into new roles.
4. **Anticipatory Socialization:** Occurs when individuals prepare for future roles they expect to occupy, such as a student observing doctors to prepare for medical school.

Agents of Socialization:

1. **Family:** Primary agent; teaches language, basic values, and emotional bonds.
2. **School:** Teaches formal knowledge, discipline, and civic values.
3. **Peer Group:** Provides social identity, friendship, and youth culture.
4. **Mass Media:** Shapes opinions, spreads information, and influences lifestyle choices.
5. **Religion:** Instills moral values and provides a worldview.
6. **Workplace:** Teaches professional norms, work ethics, and occupational identity.

14. What is Technological society? Describe its features in third world society.

[5]

Answer: Technological Society:

A technological society is one in which technology plays a central role in shaping economic, social, cultural, and political life. Technology influences how people work, communicate, learn, and interact with one another.

Features of Technological Society in Third World Countries:

1. **Uneven Development:** Technology is concentrated in urban areas while rural regions remain underdeveloped. For example, Kathmandu has 4G internet while remote villages in Nepal lack basic electricity.
2. **Technology Transfer:** Third world countries primarily import technology from developed nations rather than creating indigenous technology. This creates dependency and high costs.
3. **Dual Economy:** coexistence of modern technology-based sectors (IT, manufacturing) alongside traditional subsistence agriculture. A software company in Bangalore coexists with farmers using ox-plows.
4. **Digital Divide:** Significant gaps in access to technology between rich and poor, urban and rural, educated and uneducated populations.
5. **Cultural Impact:** Technology introduces Western values and lifestyles, sometimes conflicting with traditional cultures. Social media influences youth to adopt global trends while elders preserve tradition.
6. **Economic Opportunities:** Technology creates new job sectors (BPO, IT outsourcing) but also displaces traditional workers through automation.
7. **Government Challenges:** Weak infrastructure, limited funding, and lack of technical expertise hinder effective technology implementation in governance and public services.

15. What are the key research methods in social science?

[5]

Answer: Key Research Methods in Social Science:

1. **Survey Research:** Collects data from a large sample using questionnaires or interviews. Useful for studying attitudes, opinions, and behaviors. Example: A national census survey to understand population demographics.
2. **Participant Observation:** The researcher immerses themselves in the group being studied, observing and participating in daily activities. Example: An anthropologist living with a tribal community to understand their customs.
3. **Case Study:** In-depth examination of a single case (person, group, event, or community). Example: Studying the 2015 Nepal earthquake's impact on a specific village.
4. **Experiment:** Manipulates variables under controlled conditions to establish cause-and-effect relationships. More common in psychology than sociology. Example: Testing whether group discussion affects decision-making.
5. **Historical Research:** Examines past events, documents, and records to understand social change. Example: Analyzing royal court records to study medieval Nepalese social structure.
6. **Content Analysis:** Systematically analyzes written, verbal, or visual communication. Example: Analyzing newspaper editorials to study media bias on political issues.
7. **Focus Group Discussion:** Gathers a small group to discuss specific topics under moderator guidance. Example: Conducting FGDs with farmers to understand their adoption of new agricultural techniques.

16. What is kinship? Discuss its types.

[5]

Answer: Kinship:

Kinship refers to the social relationships based on blood ties (consanguinity), marriage (affinity), and adoption that bind individuals together in a society. It is the fundamental organizing principle of all human societies and determines rights, obligations, roles, and statuses.

Types of Kinship:

1. By Blood (Consanguineal):

Relationships based on biological descent.

- Parents and children
- Siblings (brothers and sisters)
- Grandparents and grandchildren
- Uncles, aunts, nephews, nieces

Example: A father's brother is a consanguineal relative.

2. By Marriage (Affinal):

Relationships established through marriage.

- Husband and wife
- In-laws (father-in-law, mother-in-law, brother-in-law, sister-in-law)
- Son-in-law and daughter-in-law

Example: A wife's sister is an affinal relative (sister-in-law).

3. By Adoption (Fictive/Social):

Relationships created socially rather than biologically.

- Godparents and godchildren
- Adopted parents and children
- Ritual kinship (Miteri in Nepal — sworn brotherhood/sisterhood)
- Close family friends addressed as 'aunt' or 'uncle'

Kinship Systems:

- **Patrilineal:** Descent traced through father's line (common in Nepal)
- **Matrilineal:** Descent traced through mother's line
- **Bilateral:** Equal recognition of both paternal and maternal relatives

17. What are the factors of social and cultural changes? Discuss only three factors.

[5]

Answer: Factors of Social and Cultural Change:

Social and cultural change refers to the transformation of social institutions, behaviors, beliefs, and values over time. Several factors drive these changes.

1. Technological Factors:

Technology is one of the most powerful drivers of social change.

- The internet has transformed communication, commerce, education, and social relationships.
- Mobile phones have connected rural and urban populations, changing how families maintain long-distance relationships.
- Agricultural technology has reduced manual labor but displaced traditional farming communities.
- Medical technology has increased life expectancy, changing family structures and population demographics.

2. Economic Factors:

Economic conditions significantly influence social structures and cultural practices.

- Industrialization shifted societies from agrarian to urban, changing family structures from extended to nuclear.
- Globalization has introduced foreign products, ideas, and lifestyles into local cultures.
- Economic recession can increase crime rates, mental health issues, and social unrest.
- Remittance economy in Nepal has changed rural consumption patterns and reduced agricultural labor.

3. Cultural Contact and Diffusion:

Contact between different cultures leads to exchange and modification of cultural elements.

- Tourism introduces foreign cultures to Nepal while exposing Nepalese culture to the world.
- Migration creates multicultural societies where traditions blend and evolve.
- Mass media spreads global pop culture, influencing dress, music, and language among youth.
- Education systems often promote Western scientific thinking alongside traditional knowledge systems.

Group C

Attempt any TWO questions. [2×10=20]

18. What is national integration? Discuss its major dimensions.

[10]

Answer: National Integration:

National integration is the process of unifying diverse social, cultural, religious, and ethnic groups within a nation into a cohesive whole. It creates a sense of common identity, shared destiny, and mutual loyalty among citizens, transcending regional, linguistic, and communal differences.

Major Dimensions of National Integration:

1. Political Integration:

Unifying political institutions and loyalties across the nation. This includes:

- A single constitution and legal framework applicable to all citizens
- Strong democratic institutions that represent all regions
- Federal or unitary structures that balance local autonomy with national unity
- Equal political rights and opportunities for all citizens regardless of background

2. Cultural Integration:

Creating shared cultural values while respecting diversity:

- National symbols (flag, anthem, emblem) that inspire collective pride
- National languages that facilitate communication across regions
- Shared historical narratives and national heroes
- Common festivals and celebrations that bring people together

3. Economic Integration:

Reducing economic disparities between regions and groups:

- Balanced regional development policies
- Equal access to employment and economic opportunities
- Infrastructure connecting all parts of the nation
- Equitable distribution of national resources

4. Social Integration:

Breaking down barriers between different communities:

- Inter-caste and inter-religious marriages and friendships
- Mixed educational institutions where children from all backgrounds study together
- Social mobility based on merit rather than birth
- Reduction of discrimination and prejudice

5. Emotional Integration:

Developing psychological attachment to the nation:

- Patriotism and love for the country
- Willingness to sacrifice personal interests for national good
- Trust in national institutions
- Feeling of belonging to the national community

19. Discuss the accountability of computer professionals towards society with examples.

[10]

Answer: Accountability of Computer Professionals Towards Society:

Computer professionals have significant power to shape society through the systems they design, maintain, and manage. With this power comes ethical responsibility and accountability.

1. Responsibility for System Reliability and Safety:

Computer professionals must ensure that systems function correctly and safely.

- Example: Software engineers working on aircraft control systems must rigorously test code to prevent fatal crashes. The Therac-25 radiation therapy machine incidents (1985-1987) where software bugs caused patient deaths demonstrate the consequences of inadequate testing.

2. Protection of User Privacy:

Professionals must safeguard personal data collected by systems.

- Example: Facebook's Cambridge Analytica scandal (2018) showed how failure to protect user data could manipulate elections and violate privacy. Computer professionals should implement strong encryption, access controls, and data minimization practices.

3. Ensuring Security:

Protecting systems from unauthorized access and cyberattacks.

- Example: The 2017 WannaCry ransomware attack exploited unpatched Windows systems, affecting hospitals worldwide. IT professionals have a duty to promptly apply security patches and maintain robust defenses.

4. Transparency and Honesty:

Being truthful about system capabilities, limitations, and risks.

- Example: AI developers should disclose when algorithms make decisions that affect people's lives (loan approvals, hiring, criminal sentencing) and acknowledge algorithmic biases.

5. Accessibility and Digital Inclusion:

Designing systems that are accessible to people with disabilities and those with limited resources.

- Example: Web developers should follow WCAG guidelines to ensure websites work with screen readers for visually impaired users.

6. Environmental Responsibility:

Considering the environmental impact of computing infrastructure.

- Example: Data centers consume enormous electricity. Professionals should optimize code efficiency, use cloud resources responsibly, and advocate for renewable energy.

7. Professional Codes of Ethics:

Organizations like ACM and IEEE have established codes of ethics that guide professional conduct, emphasizing public interest, honesty, and respect for intellectual property.

20. What are the basic steps of proposal writing in social research? Describe in brief.

[10]

Answer: Basic Steps of Proposal Writing in Social Research:

A research proposal is a detailed plan that outlines what the researcher intends to investigate, why it is important, and how the research will be conducted.

1. Title Page:

Includes the research title, researcher's name, institutional affiliation, and date. The title should be concise, descriptive, and capture the essence of the study.

2. Introduction/Background:

Provides context for the research problem. Explains why the topic is important, its relevance to society, and what gap in existing knowledge the research aims to fill.

3. Statement of the Problem:

Clearly defines the specific issue or question the research will address. A well-formulated problem statement is specific, researchable, and significant.

4. Research Objectives:

Lists the specific goals the research aims to achieve. Objectives should be SMART (Specific, Measurable, Achievable, Relevant, Time-bound).

5. Literature Review:

Surveys existing research on the topic to identify gaps, establish theoretical framework, and show how the proposed research contributes to knowledge.

6. Theoretical Framework:

Identifies the theories, concepts, and models that will guide the research and interpretation of findings.

7. Research Questions/Hypotheses:

Formulates specific questions to be answered or hypotheses to be tested based on the theoretical framework.

8. Methodology:

Details how the research will be conducted:

- Research design (qualitative, quantitative, or mixed)
- Population and sampling method
- Data collection methods (surveys, interviews, observation)
- Data analysis techniques
- Ethical considerations

9. Work Plan/Time Schedule:

Presents a timeline showing when each phase of the research will be completed.

10. Budget:

Estimates the financial resources needed for the research, including personnel, equipment, travel, and miscellaneous expenses.

11. Expected Outcomes:

Describes what the research hopes to discover and how the findings will benefit society, policy, or theory.

12. References/Bibliography:

Lists all sources cited in the proposal using a consistent citation style (APA, MLA, Chicago).

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.